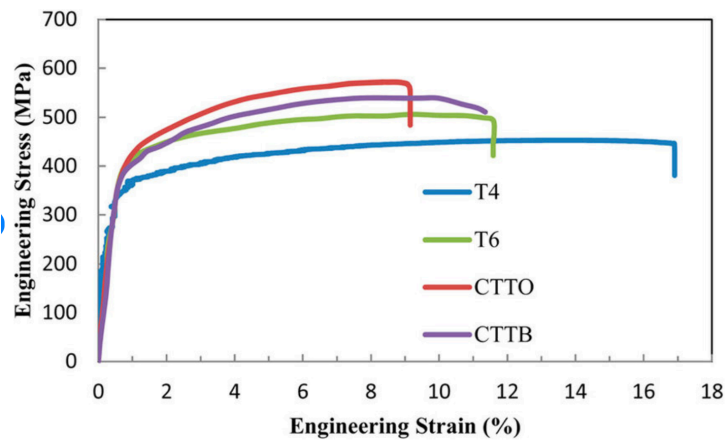
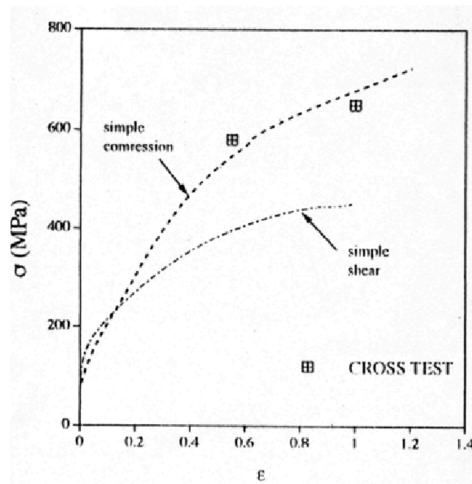


Alloy class	Example alloy & condition	Main strengthening methods	Roles of main alloying elements	Typical shapes & processes	Weldability	Typical engineering applications & required properties	Outstanding property	Main weaknesses / common failures	Useful surface treatments	Main sources
Al- based	2024-T4 Al-Cu-Mg	Precipitation hardening; some solid-solution and work hardening.	Cu forms tiny strengthening precipitates. Mg helps precipitation and adds strength. Mn/Fe helps grain control.	Rolled sheet/plate, extrusions solution heat treated and quenched.	Poor fusion weldability and risk of cracking and strength loss.	Aircraft fuselage and wing skins - weight to strength ratio and fatigue resistance.	Very high strength-to-weight for riveted airframe structures.	Corrosion and stress-corrosion cracking	anodizing, paint	https://www.engineeringnotes.com/industrial-engineering/metals/list-of-metals-and-their-alloys-material-engineering/34168
Cu- based	70-30 brass	Solid-solution strengthening of Cu by Zn and cold working	Zn increases strength and hardness while keeping good ductility allowing forming	Strip, sheet, tube. Made by rolling and deep drawing.	Good brazing and soldering.	Cartridge cases, radiator and heat-exchanger tubes - easily formable and good thermal conductivity and corrosion resistance in water.	Cracking in ammonia environments	Cracking in ammonia environments. Ammonia	Protective coating, tin plating	https://www.munotmetalloys.com/blog/choosing-the-right-brass-70-30-tubes-for-industrial-applications/
Carbon-steel-based	AISI 4135 Cr-Mo steel	Transformation hardening plus tempering.	C controls hardness and martensite formation. Cr and Mo increase hardness, improve strength and toughness	Bars, shafts, gears. Forged or rolled, machined.	Moderate - needs to be preheated and heat treated after welding to avoid cracking.	Automotive axles, gears and shafts - high strength, fatigue resistance and toughness.		Fatigue cracking	Carburizing or nitriding	https://asm.matweb.com/search/specificmaterial.asp?bassnum=mq304a
Stainless-steel-based	AISI 304 stainless steel	Solid-solution strengthening from Cr and Ni and work hardening.	Cr helps corrosion resistance and adds strength. Ni stabilizes it and improves toughness and formability.	Sheet, plate, tube, bar. Produced by rolling and forming, cold-worked to increase strength.	Good weldability by TIG or MIG welding.	Food processing equipment, chemical tanks, kitchenware.- Excellent corrosion resistance, toughness and easily formable.		Corrosion after improper welding or high-temperature exposure.		https://metalzenith.com/blogs/steel-properties/4135-steel-properties-and-key-applications https://premiumalu.com/2024-t4-aluminum-properties/



https://www.researchgate.net/figure/The-engineering-stress-strain-curves-of-2024-aluminium-alloys-for-different-treatments_fig3_321581599



https://www.researchgate.net/figure/Equivalent-stress-strain-response-of-7030-brass-in-simple-shear-and-simple-compression_fig7_237275930

Sample	Young's modulus (GPa)	0.2% Yield stress (MPa)	Ultimate tensile strength (MPa)	Elongation (%)
Sample A	205	1030	1170	9.5
Sample B	200	1020	1180	6.6
Sample C	202	1020	1140	11.3
Average	202.3	1023	1163	9.1

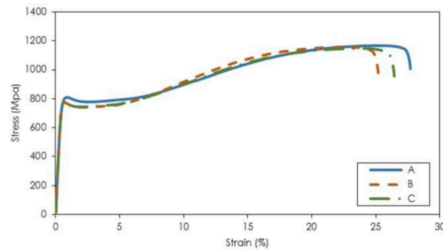


Figure 4 Stress-strain curves of 300 µm thickness 304 stainless steel sheet

https://www.researchgate.net/figure/Tensile-properties-of-100-m-thickness-304-stainless-steel-sheet_tbl1_305405777